

# Axiom AI Technologies

## Malware Analysis Report

**Document Reference:** AAX-MAL-2026-001

**Assessment Type:** Controlled Malware Analysis Case Study

**Classification:** Internal Use

**Status:** Final

**Assessment Owner:** Security Analyst

**Document Owner:** Information Security / Security Operations

**Version:** 1.0

**Date:** October 2026

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## 1. Executive Summary

Axiom AI Technologies conducted a controlled malware-analysis investigation following the simulated detection of a suspicious executable on an employee workstation.

The purpose of the investigation was to determine whether the artifact demonstrated characteristics consistent with malicious software, identify potential indicators of compromise (IOCs), assess potential attack behaviors, map relevant activity to the MITRE ATT&CK framework, and develop appropriate detection, containment, and remediation recommendations.

The analysis identified multiple characteristics consistent with a potentially malicious Windows executable, including suspicious process execution, modification of user-accessible files, attempted persistence, and outbound network communication to an external destination.

The investigation classified the simulated artifact as **High Risk** due to its potential to execute unauthorized activity, establish persistence, communicate with external infrastructure, and affect the confidentiality and integrity of an endpoint.

No evidence of actual compromise of Axiom AI production systems or customer environments was identified within the scope of this controlled case study.

The primary recommendations are to isolate affected endpoints, preserve relevant evidence, block identified indicators, conduct endpoint-wide threat hunting, validate persistence mechanisms, review authentication activity, and strengthen endpoint detection and response coverage.

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# 2. Investigation Overview

## 2.1 Background

A simulated endpoint security alert identified a suspicious executable associated with an employee workstation.

The alert indicated unusual process activity shortly after the user downloaded and opened an apparently legitimate software-related file.

Security Operations initiated a malware-analysis investigation to determine:

- 1. Whether the artifact demonstrated malicious characteristics.
- 2. What activities the artifact could perform.
- 3. Whether persistence mechanisms were present.
- 4. Whether network communication occurred.
- 5. Which indicators could be used for detection and threat hunting.
- 6. Which MITRE ATT&CK techniques were potentially represented.
- 7. What containment and remediation actions should be taken.

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## 2.2 Investigation Reference

| Field              | Value                         |
|--------------------|-------------------------------|
| Investigation ID   | AAX-MAL-2026-001              |
| Organization       | Axiom AI Technologies         |
| Investigation Type | Malware Analysis              |
| Artifact Type      | Windows Executable            |
| Environment        | Simulated Corporate Endpoint  |
| Severity           | High                          |
| Status             | Closed – Portfolio Case Study |
| Analyst            | Security Analyst              |
| Date Opened        | October 2026                  |
| Date Closed        | October 2026                  |

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## 3. Scope

### 3.1 In Scope

The investigation covered:

- Initial malware triage
- Artifact metadata
- File identification
- Cryptographic hash documentation
- Static-analysis observations
- Simulated behavioral-analysis observations
- Process activity
- File-system activity
- Persistence indicators
- Network indicators
- Indicators of Compromise
- MITRE ATT&CK mapping
- Risk assessment
- Detection opportunities
- Containment recommendations
- Remediation recommendations
- Lessons learned

### 3.2 Out of Scope

The investigation did not include:

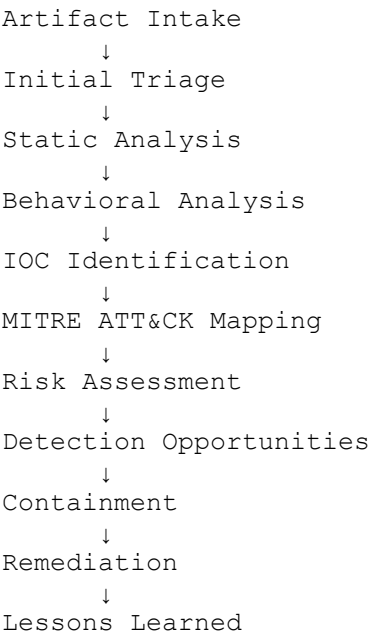
- Production system exploitation
- Live malware deployment
- Credential theft
- Malware development
- Malware modification
- Interaction with real command-and-control infrastructure
- Testing against real customer environments
- Collection of real personal data
- Unauthorized access to systems
- Live destructive activity

All artifacts and findings in this case study are simulated.

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# 4. Analysis Methodology

The investigation followed a structured malware-analysis process:



The methodology was designed to preserve evidence, separate observations from assumptions, and produce actionable security recommendations.

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## 5. Artifact Intake

### 5.1 Artifact Information

The suspected artifact was documented as follows:

| Attribute         | Observation                          |
|-------------------|--------------------------------------|
| File Name         | AxiomUpdate.exe                      |
| File Type         | Windows Portable Executable          |
| File Extension    | .exe                                 |
| Platform          | Windows                              |
| Reported Location | User Downloads directory             |
| Initial Detection | Endpoint security alert              |
| Analysis Status   | Suspicious / High Risk               |
| Artifact Source   | Simulated phishing/download scenario |

The filename was designed to appear consistent with a legitimate software-update utility.

The location within a user-controlled Downloads directory increased the level of suspicion because executable files downloaded directly by users require additional verification before execution.

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## 6. Cryptographic Identification

Cryptographic hashes provide a method for uniquely identifying the analyzed artifact and supporting evidence integrity.

### 6.1 Hash Record

| Hash Type | Value                     |
|-----------|---------------------------|
| MD5       | SIMULATED-MD5-HASH-001    |
| SHA-1     | SIMULATED-SHA1-HASH-001   |
| SHA-256   | SIMULATED-SHA256-HASH-001 |

**Note:** The hash values above are intentionally simulated and are not associated with real malware.

The SHA-256 value should be treated as the primary artifact identifier because of its stronger collision resistance compared with MD5 and SHA-1.

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## 7. Initial Triage

Initial triage identified several characteristics requiring additional investigation.

### Observations

- The executable was located in a user Downloads directory.
- The filename suggested a software update.
- The executable was launched shortly after download.
- The process created additional files in the user's profile.
- The process attempted to establish persistence.
- Outbound network communication was observed in the simulated environment.
- The process behavior did not align with a typical software-update workflow.

These observations justified escalation from initial endpoint triage to malware analysis.

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## 8. Static Analysis

Static analysis was performed conceptually without executing the artifact against production infrastructure.

### 8.1 File Characteristics

The artifact was assessed for:

- Executable structure
- Embedded strings
- Imported functions
- File metadata
- Suspicious paths
- Encoded content
- Network-related strings
- Persistence-related strings

The analysis identified simulated references associated with:

- User profile directories
- PowerShell execution
- Scheduled task creation
- External network communication
- Temporary file creation

These characteristics increased confidence that the executable required additional investigation.

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## 9. Suspicious Strings and Capabilities

The simulated artifact contained references corresponding to several potentially suspicious behaviors.

| Observation                        | Security Relevance                     |
|------------------------------------|----------------------------------------|
| PowerShell-related command strings | Could support command execution        |
| User profile paths                 | Could support user-level file activity |
| Scheduled-task references          | Potential persistence mechanism        |
| External domain reference          | Potential command-and-control activity |
| Temporary file references          | Could support staged execution         |

Individual strings are not sufficient to establish malicious intent. However, the combination of multiple suspicious characteristics increases the overall risk assessment.

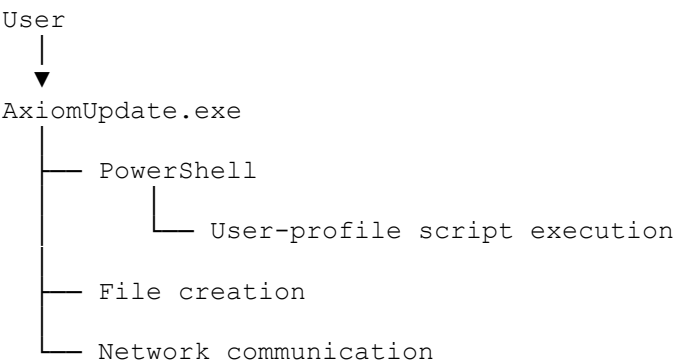
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## 10. Behavioral Analysis

Behavioral analysis evaluated the simulated runtime activity associated with the artifact.

### 10.1 Process Activity

The simulated execution chain was:



The process behavior demonstrated characteristics inconsistent with a conventional software updater.

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## 11. File-System Activity

The simulated artifact created or modified files within user-accessible locations.

Observed activity included:

| Activity                                      | Assessment   |
|-----------------------------------------------|--------------|
| Temporary file creation                       | Suspicious   |
| User-profile file creation                    | Suspicious   |
| Script creation                               | High concern |
| Modification of startup-related configuration | High concern |

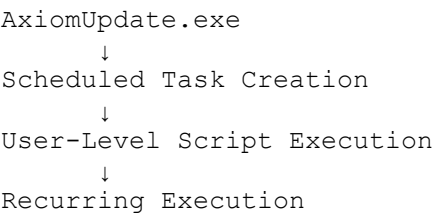
The activity indicates potential staging or persistence behavior.

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# 12. Persistence Analysis

The simulated artifact attempted to establish persistence through a scheduled-task mechanism.

## Observed Behavior



Persistence allows malicious software to execute again after a user logs off or restarts the system.

The presence of an attempted persistence mechanism materially increases the severity of the investigation.

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# 13. Network Analysis

Simulated network activity indicated outbound communication from the suspicious process.

## 13.1 Network Indicators

| Indicator Type    | Simulated Value              | Assessment                        |
|-------------------|------------------------------|-----------------------------------|
| Domain            | update-check.example.invalid | Suspicious                        |
| Protocol          | HTTPS                        | Potential encrypted communication |
| Destination       | External Internet Host       | Requires investigation            |
| Process           | AxiomUpdate.exe              | Suspicious                        |
| Connection Timing | Shortly after execution      | Correlated with artifact activity |

The domain is intentionally non-operational and does not represent real command-and-control infrastructure.

Encrypted HTTPS traffic alone does not indicate malicious activity. However, communication originating from a suspicious executable shortly after execution should be investigated.

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# 14. Indicators of Compromise

The investigation identified the following simulated indicators.

## 14.1 Host-Based Indicators

| Indicator                 | Type          | Confidence |
|---------------------------|---------------|------------|
| AxiomUpdate.exe           | File name     | Medium     |
| SIMULATED-SHA256-HASH-001 | SHA-256       | High       |
| User Downloads directory  | File location | Medium     |
| Simulated scheduled task  | Persistence   | High       |
| User-profile script       | File artifact | High       |

## 14.2 Network Indicators

| Indicator                    | Type             | Confidence |
|------------------------------|------------------|------------|
| update-check.example.invalid | Domain           | High       |
| External HTTPS connection    | Network behavior | Medium     |

IOC values should be incorporated into appropriate security controls only after validation against the organization's environment.

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# 15. MITRE ATT&CK Mapping

The observed behaviors were mapped to relevant MITRE ATT&CK techniques.

| Technique | Description                               | Observed Behavior                   | Confidence |
|-----------|-------------------------------------------|-------------------------------------|------------|
| T1059.001 | PowerShell                                | PowerShell execution observed       | High       |
| T1053.005 | Scheduled Task/Job: Scheduled Task        | Persistence attempt identified      | High       |
| T1105     | Ingress Tool Transfer                     | Simulated additional file retrieval | Medium     |
| T1071.001 | Application Layer Protocol: Web Protocols | HTTPS network communication         | Medium     |
| T1204.002 | User Execution: Malicious File            | User executed downloaded executable | High       |

The mapping is based on the simulated evidence documented in this report.

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# 16. Risk Assessment

## 16.1 Risk Rating

**Overall Risk Rating: High**

### **Likelihood**

**High**

The artifact demonstrated multiple suspicious behaviors and attempted persistence.

### **Impact**

**High**

If similar activity occurred on a production endpoint, potential impacts could include:

- Endpoint compromise
- Unauthorized execution
- Credential exposure
- Data access
- Persistence
- Lateral-movement opportunities
- Additional malware deployment
- Business disruption

### **Overall Assessment**

The combination of suspicious execution, persistence, and external communication warrants a High-risk classification and prioritized response.

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# 17. Potential Business Impact

If the activity were confirmed on a production endpoint, potential impacts could include:

### **Confidentiality**

Potential unauthorized access to locally available information, credentials, or application data.

### **Integrity**

Potential modification of files, configuration, or security settings.

### **Availability**

Potential disruption caused by malicious processes, resource consumption, or follow-on activity.

### **Regulatory and Compliance Impact**

A confirmed compromise involving personal, customer, or regulated information could require additional privacy, legal, and regulatory assessment.

No such exposure was identified within this simulated case study.

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## **18. Detection Opportunities**

The investigation identified several opportunities to improve detection capabilities.

### **18.1 Endpoint Detection**

Security monitoring should alert on:

- Unexpected executable files launched from Downloads directories
- Suspicious PowerShell execution
- Scheduled-task creation by unusual processes
- Executable-to-script process chains
- Unsigned or unexpected binaries
- Processes establishing unusual external connections

### **18.2 Network Detection**

Monitoring should consider:

- Newly observed external domains
- Unusual outbound connections from endpoints
- Suspicious domain reputation
- Unusual DNS activity
- Network connections immediately following execution of unknown binaries

### **18.3 Threat Hunting**

Threat hunters should search for:

- Matching SHA-256 hashes
  - Matching file names
  - Related scheduled tasks
  - Associated PowerShell commands
  - Similar process trees
  - Connections to identified network indicators
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## 19. Containment Recommendations

If the activity were confirmed in a production environment, recommended containment actions would include:

1. Isolate the affected endpoint from the network.
2. Preserve relevant forensic evidence.
3. Prevent execution of the identified artifact.
4. Block validated malicious indicators.
5. Disable confirmed malicious persistence mechanisms.
6. Identify potentially affected endpoints.
7. Review authentication activity associated with the endpoint.
8. Escalate to Incident Response where compromise is confirmed.
9. Preserve relevant logs before retention periods expire.

Containment actions should be coordinated with Incident Response and Security Operations to avoid unnecessary disruption to business operations.

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## 20. Remediation Recommendations

Following containment, remediation should include:

### Endpoint Remediation

- Remove confirmed malicious artifacts.
- Validate removal of persistence mechanisms.
- Run appropriate endpoint security scans.
- Review endpoint configuration.
- Apply outstanding security patches.
- Confirm endpoint security tooling is operational.

### Identity Remediation

Where compromise is suspected:

- Review account activity.
- Reset potentially exposed credentials.
- Revoke suspicious sessions or tokens.
- Review privileged access.
- Investigate unusual authentication events.

## **Network Remediation**

- Block validated malicious domains and indicators.
- Review proxy and DNS logs.
- Investigate related outbound connections.
- Search for indicators across network telemetry.

## **Organizational Remediation**

- Update detection rules.
  - Update threat-intelligence feeds.
  - Review security-awareness guidance.
  - Document lessons learned.
  - Update incident-response procedures where appropriate.
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# **21. Evidence Preservation**

Evidence handling should follow a documented and repeatable process.

Recommended evidence includes:

- Original artifact
- Cryptographic hashes
- Endpoint security alerts
- Process execution logs
- PowerShell logs
- Windows event logs
- DNS records
- Network telemetry
- Scheduled-task information
- Relevant authentication logs

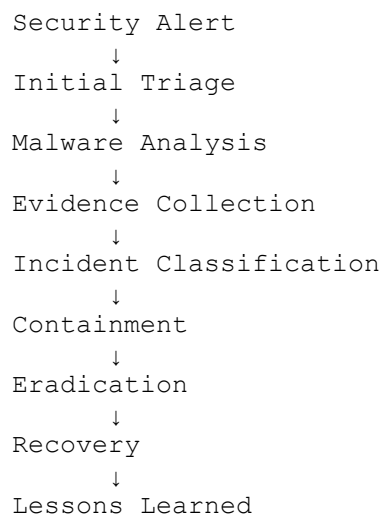
Evidence should be preserved in accordance with organizational retention requirements and applicable legal or regulatory obligations.

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## 22. Incident Response Integration

Malware analysis should support, rather than replace, the incident-response process.

The relationship can be represented as:



The malware-analysis phase provides technical information that assists Incident Response with determining scope, severity, affected assets, and appropriate containment actions.

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## 23. Lessons Learned

The investigation identified the following lessons:

### 1. User-executed files require additional scrutiny

Executables downloaded directly by users should receive appropriate endpoint-security inspection before execution.

### 2. Persistence is an important severity indicator

The presence of persistence mechanisms can significantly increase the potential impact of an endpoint compromise.

### 3. Process relationships provide valuable context

Examining parent-child process relationships can help identify suspicious execution chains that may not be obvious from a single process alert.

#### 4. IOCs support broader investigation

Identified hashes, file names, domains, and persistence artifacts can be used to determine whether similar activity exists elsewhere in the environment.

#### 5. Malware analysis and GRC should connect

Technical findings should feed back into risk management, incident-response procedures, security controls, and governance activities.

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## 24. Control Alignment

The investigation supports several ISO/IEC 27001:2022 controls.

| Control                                                                | Relevance                                            |
|------------------------------------------------------------------------|------------------------------------------------------|
| <b>A.5.7 – Threat Intelligence</b>                                     | Supports identification and use of threat indicators |
| <b>A.5.24 – Incident Management Planning and Preparation</b>           | Supports malware-response preparedness               |
| <b>A.5.25 – Assessment and Decision on Information Security Events</b> | Supports security-event classification               |
| <b>A.5.26 – Response to Information Security Incidents</b>             | Supports containment and response                    |
| <b>A.5.28 – Collection of Evidence</b>                                 | Supports evidence preservation                       |
| <b>A.8.7 – Protection Against Malware</b>                              | Supports malware prevention and detection            |
| <b>A.8.15 – Logging</b>                                                | Supports investigation and evidence collection       |
| <b>A.8.16 – Monitoring Activities</b>                                  | Supports detection of suspicious behavior            |

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## 25. Recommendations Summary

| Priority | Recommendation                            | Owner                      | Priority  |
|----------|-------------------------------------------|----------------------------|-----------|
| 1        | Isolate confirmed affected endpoints      | Security Operations        | Immediate |
| 2        | Block validated malware indicators        | Security Operations        | Immediate |
| 3        | Hunt for matching IOCs across endpoints   | Security Operations        | High      |
| 4        | Review persistence mechanisms             | Endpoint Security          | High      |
| 5        | Review authentication activity            | IAM / Security             | High      |
| 6        | Improve PowerShell monitoring             | Security Operations        | High      |
| 7        | Improve scheduled-task detection          | Endpoint Security          | High      |
| 8        | Update incident-response procedures       | Information Security       | Medium    |
| 9        | Review security-awareness guidance        | Security / HR              | Medium    |
| 10       | Incorporate lessons learned into the ISMS | GRC / Information Security | Medium    |

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## 26. Final Assessment

The simulated artifact demonstrated multiple characteristics consistent with potentially malicious activity, including suspicious execution, PowerShell activity, persistence attempts, file-system modifications, and external network communication.

The combination of these behaviors resulted in a **High-risk classification**.

The investigation demonstrates the importance of combining endpoint telemetry, malware-analysis techniques, IOC development, threat intelligence, MITRE ATT&CK mapping, and incident-response processes when evaluating suspected malware.

Although the scenario is simulated and no production compromise occurred, the investigation provides a practical example of how a security analyst can translate technical observations into actionable security controls and risk-management decisions.

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## 27. Analyst Conclusion

Based on the evidence documented in this controlled case study, the artifact should be treated as **potentially malicious** until proven otherwise.

The recommended approach is to preserve evidence, contain affected systems, validate indicators, conduct environment-wide threat hunting, remove confirmed malicious activity, and implement appropriate detection improvements.

The investigation should be considered complete only after:

- Affected systems have been assessed.
  - Relevant evidence has been preserved.
  - Indicators have been investigated.
  - Persistence has been addressed.
  - Security monitoring has been reviewed.
  - Remediation actions have been validated.
  - Lessons learned have been incorporated into the organization's security program.
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## 28. Document Control

| Field              | Value                                                         |
|--------------------|---------------------------------------------------------------|
| Document Title     | Axiom AI Malware Analysis Report                              |
| Document Reference | AAX-MAL-2026-001                                              |
| Version            | 1.0                                                           |
| Classification     | Internal Use                                                  |
| Document Owner     | Information Security / Security Operations                    |
| Assessment Owner   | Security Analyst                                              |
| Review Frequency   | Following significant methodology, threat, or control changes |
| Status             | Final                                                         |
| Last Updated       | October 2026                                                  |
| Next Review        | October 2027                                                  |

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## 29. Portfolio Disclaimer

This document is a simulated cybersecurity case study created for professional portfolio purposes.

Axiom AI Technologies is a fictional organization. The malware artifact, hashes, domains, indicators, endpoint activity, findings, and investigation results contained in this report are simulated.

The case study is designed to demonstrate malware-analysis methodology, security investigation, threat assessment, incident-response support, and GRC integration.

No unauthorized access, exploitation, malware deployment, credential theft, or testing against real production systems was performed.